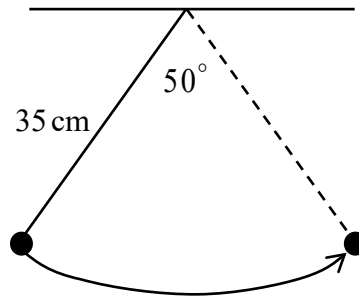


Question 1

T1

A pendulum that is 35 cm long swings through an angle of 50° . Determine the length of the arc through which the pendulum swings.



Solution

$$\theta = 50 \left(\frac{\pi}{180} \right)$$

1 mark for conversion

$$= \frac{5\pi}{18}$$

or

$$= 0.872\ 664\dots$$

$$s = \theta r$$

$$= \left(\frac{5\pi}{18} \right) (35)$$

1 mark for substitution

$$= \frac{175\pi}{18} \text{ cm}$$

2 marks

or

$$= 30.543 \text{ cm}$$

Solve algebraically, where $0 \leq \theta \leq 2\pi$.

$$2 \cos^2 \theta = \sin^2 \theta - 2 \cos \theta$$

Solution

$$2 \cos^2 \theta = 1 - \cos^2 \theta - 2 \cos \theta$$

1 mark for substitution of appropriate identity

$$3 \cos^2 \theta + 2 \cos \theta - 1 = 0$$

$$(\cos \theta + 1)(3 \cos \theta - 1) = 0$$

$$\cos \theta = -1 \quad \cos \theta = \frac{1}{3}$$

1 mark for solving for $\cos \theta$

$$\theta = \pi \quad \theta_r = 1.230\,959\dots$$

$$\theta = 1.231, 5.052$$

2 marks for solving for θ (1 mark for each branch)

4 marks

Determine the number of arrangements of the letters in the word ATTENTION which begin with the letter A.

Solution

$$\frac{8!}{3!2!} = 3360$$

1 mark for 8!

1 mark for division by 3!2! ($\frac{1}{2}$ mark for 3!; $\frac{1}{2}$ mark for 2!)

2 marks

Solve for x , algebraically.

$$e^{2x+1} = 5^x$$

Solution

$$\ln(e^{2x+1}) = \ln(5^x)$$

½ mark for applying logarithms

$$(2x+1)\ln e = x\ln 5$$

1 mark for power law (½ mark for each)

$$2x+1 = x\ln 5$$

$$2x - x\ln 5 = -1$$

½ mark for collecting terms with x

$$x(2 - \ln 5) = -1$$

$$x = \frac{-1}{2 - \ln 5}$$

½ mark for isolating x

$$x = -2.560\ 412\dots$$

$$x = -2.560$$

½ mark for evaluating quotient of logarithms

3 marks

There are 10 teachers and 17 students who would like to attend a field trip.

Determine the number of ways that 3 teachers and 9 students can be selected given that Mr. Jones and Mrs. Carol, two of the teachers, must be selected to attend the field trip.

Solution

$${}_2C_2 \cdot {}_8C_1 \cdot {}_{17}C_9$$

$$194\,480$$

1 mark for ${}_8C_1$

$\frac{1}{2}$ mark for ${}_{17}C_9$

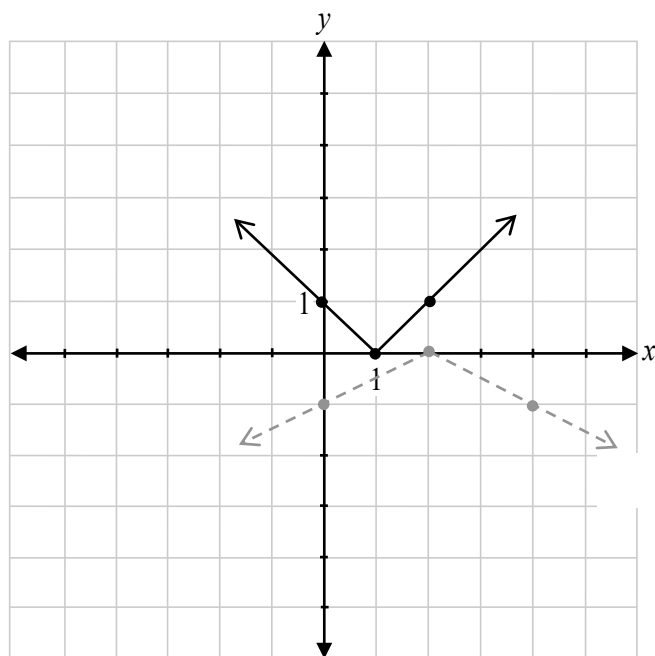
$\frac{1}{2}$ mark for product of combinations

2 marks

Note:

- ${}_2C_2$ does not need to be shown.

Given the graph of $f(x)$, sketch the graph of $y = -f(2x)$.

Solution

1 mark for vertical reflection
1 mark for horizontal compression

2 marks

There are 5 roads between Anneville and Berrybourg, and 2 roads between Berrybourg and Carriton.

Determine how many ways Blake can travel from Anneville to Carriton and back to Anneville, given the following conditions:

- he must travel through Berrybourg in both directions
- he cannot use the same road twice

Solution

$$\frac{5}{A \rightarrow B} \cdot \frac{2}{B \rightarrow C} \cdot \frac{1}{C \rightarrow B} \cdot \frac{4}{B \rightarrow A}$$

40 ways

$\frac{1}{2}$ mark for correct number of routes between towns
 $\frac{1}{2}$ mark for product

1 mark

Determine which term contains x^0 in the binomial expansion of $\left(x^2 + \frac{1}{x}\right)^6$.

Solution

Method 1

$$x^0 = \left(x^2\right)^{6-k} \left(\frac{1}{x}\right)^k$$

$\frac{1}{2}$ mark for substitution

$$x^0 = x^{12-2k} x^{-k}$$

$$x^0 = x^{12-3k}$$

$$3k = 12$$

$$k = 4$$

$\frac{1}{2}$ mark for solving for k

$\therefore$ the fifth term

1 mark for the fifth term (or a term consistent with the value of k)

2 marks

Method 2

$$\left(x^2\right)^6, \left(x^2\right)^5 \left(\frac{1}{x}\right), \left(x^2\right)^4 \left(\frac{1}{x}\right)^2$$

1 mark for determining the pattern

$$x^{12}, x^9, x^6 \dots$$

$\therefore$ the fifth term

1 mark for the fifth term (or a term consistent with the pattern)

2 marks

Given $f(x) = x^3 + 1$, determine the equation of $f^{-1}(x)$.

Solution

Let $y = f(x)$

$$y = x^3 + 1$$

To determine the inverse of $f(x)$, switch x and y .

$$x = y^3 + 1$$

1 mark for switching x and y

$$x - 1 = y^3$$

$$\sqrt[3]{x-1} = y$$

$\frac{1}{2}$ mark for isolating y

$$f^{-1}(x) = \sqrt[3]{x-1}$$

$\frac{1}{2}$ mark for writing equation in terms of $f^{-1}(x)$

2 marks

Guillermo was asked to determine the number of ways to select a president, a vice president, and a treasurer from a group of 11 people.

His solution: ${}_{11}C_3$.

Explain why he should have used a permutation instead of a combination.

Solution

He should have used a permutation because the people are being selected for specific positions.

1 mark

Prove the following identity for all permissible values of x .

$$\frac{\csc^2 x \sec x}{\tan x + \cot x} = \csc x$$

Solution

Left-Hand Side	Right-Hand Side
$\frac{\left(\frac{1}{\sin^2 x}\right)\left(\frac{1}{\cos x}\right)}{\frac{\sin x}{\cos x} + \frac{\cos x}{\sin x}}$ $\frac{\frac{1}{\sin^2 x \cos x}}{\frac{\sin^2 x + \cos^2 x}{\sin x \cos x}}$ $\frac{1}{\sin^2 x \cos x} \cdot \frac{\sin x \cos x}{1}$ $\frac{1}{\sin x}$ $\csc x$	$\csc x$

1 mark for correct substitution of identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Determine the value of x , algebraically.

$$5\log_a 2 - \frac{1}{4}\log_a 16 = \log_a x$$

Solution

$$\log_a 2^5 - \log_a 16^{\frac{1}{4}} = \log_a x$$

1 mark for power law ($\frac{1}{2}$ mark for each)

$$\log_a \left(\frac{32}{2} \right) = \log_a x$$

1 mark for quotient law

$$\log_a 16 = \log_a x$$

$$x = 16$$

1 mark for equating arguments

3 marks

Tamara must determine the factors of $x^4 - 13x + 2x^3 - 14x^2 + 24$.

Explain why the coefficients Tamara used to set up her synthetic division are not written correctly.

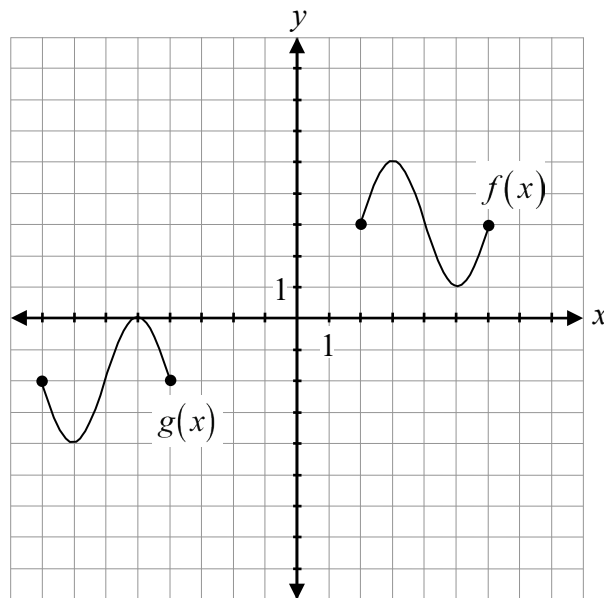
Handwritten synthetic division setup for the polynomial $x^4 - 13x + 2x^3 - 14x^2 + 24$. The coefficients are written as 1, -13, 2, -14, 24, separated by a vertical line.

Solution

Tamara did not arrange the coefficients of the terms in order of descending degree.

1 mark

Determine the equation of the graph of $g(x)$ in terms of $f(x)$.



Solution

$$g(x) = \underline{f(-(x+2)) - 5}$$

1 mark for horizontal reflection
1 mark for horizontal translation
1 mark for vertical translation

or

$$g(x) = \underline{-f(x+10) + 1}$$

1 mark for vertical reflection
1 mark for horizontal translation
1 mark for vertical translation

3 marks

Expand, using the laws of logarithms.

$$\log_2 \left[\frac{(x-1)(x-2)}{x} \right]$$

Solution

$$\log_2 (x-1) + \log_2 (x-2) - \log_2 x$$

1 mark for product law
1 mark for quotient law

2 marks

Question 16**R3**

Identify the range of the function $g(x) = \frac{1}{2}f(x+1)$, given that the range of the function $y = f(x)$ is $[-6, 4]$.

a) $[-12, 8]$

b) $[-7, 3]$

c) $[-5, 5]$

d) $[-3, 2]$

Question 17**P4**

Identify the value of a , given that there are 11 terms in the expansion of $(3x^4 - y)^{2a}$.

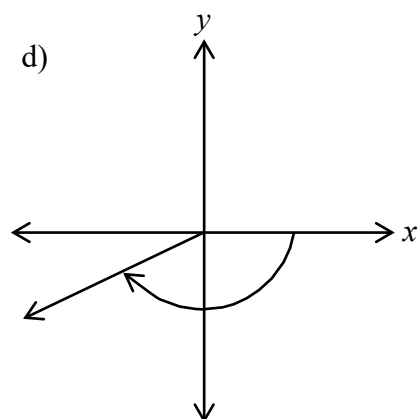
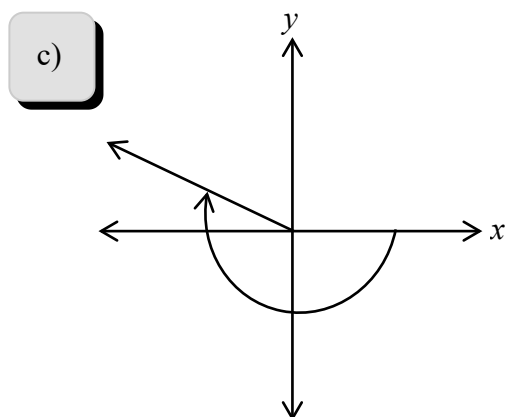
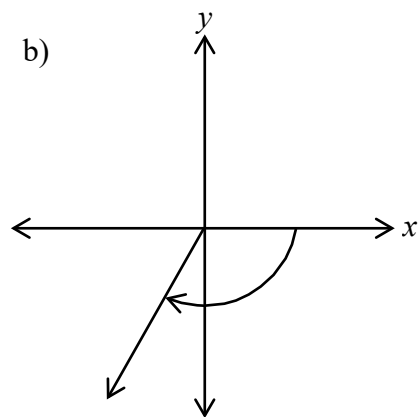
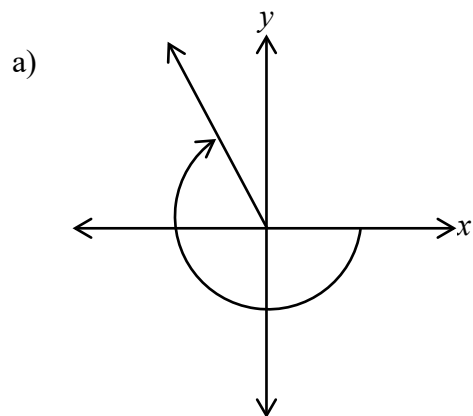
a) 5

b) 6

c) 10

d) 11

Identify the angle that best represents $\theta = -\frac{6\pi}{5}$.



Question 19

R12

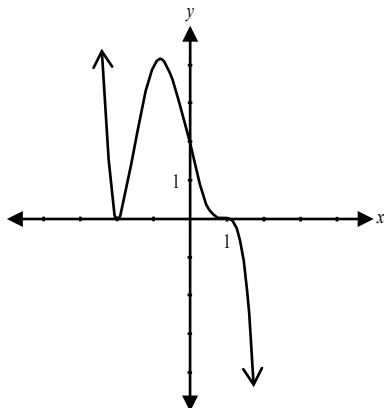
Identify a possible value for n , given the graph of $y = -\frac{1}{2}(x+2)^2(x-1)^n$.

a) 1

b) 2

c) 3

d) 4



Question 20

R14

Identify the statement that is false, given $g(x) = \frac{8x^2}{x^2 - 16}$.

a) the graph of $g(x)$ has one x -intercept.b) the graph of $g(x)$ has a point of discontinuity (hole) at $x = 0$.c) the graph of $g(x)$ has two vertical asymptotes.d) the graph of $g(x)$ has a horizontal asymptote at $y = 8$.

Question 21

R7

Identify the equivalent form of $\log_a\left(\frac{1}{x^2}\right)$.

a) $-2\log_a x$ b) $1 - 2\log_a x$ c) $2\log_a x$ d) $-2\log_a\left(\frac{1}{x}\right)$

Question 22**P3**

Identify which one of the following expressions is equivalent to ${}_{13}C_6$.

a) ${}_{13}P_6$

b) ${}_{13}C_7$

c) ${}_{12}P_7$

d) ${}_{12}C_6$

Question 23**R1**

Identify the equation of $h(x) = f(x) - g(x)$, given $f(x) = x + 5$ and $g(x) = 4x + 1$.

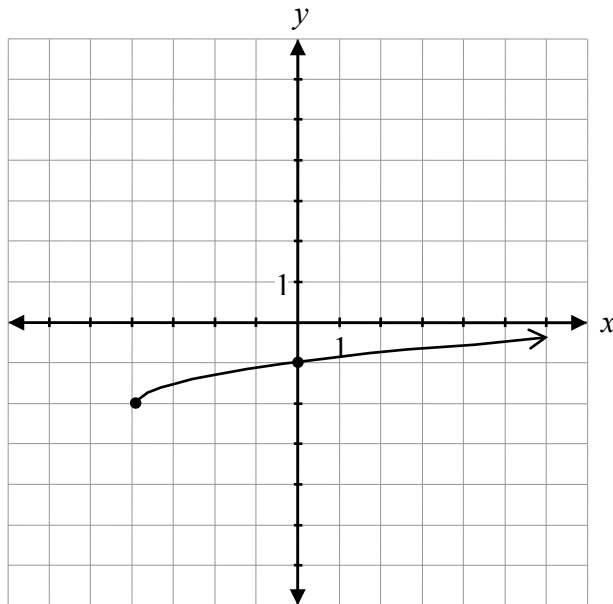
a) $h(x) = -3x + 6$

b) $h(x) = -3x + 4$

c) $h(x) = 3x + 6$

d) $h(x) = 3x - 4$

Determine the equation of the radical function represented by the graph.

**Solution**

$$y = \frac{1}{2}\sqrt{x+4} - 2$$

1 mark for vertical compression
1 mark for horizontal translation
1 mark for vertical translation

3 marks

or

$$y = \sqrt{\frac{1}{4}(x+4)} - 2$$

1 mark for horizontal stretch
1 mark for horizontal translation
1 mark for vertical translation

3 marks

Determine the exact value of x .

$$\sec\left(\frac{2\pi}{3}\right)\left(\sin\left(-\frac{5\pi}{3}\right)\right)(x) = 3$$

Solution

$$(-2)\left(\frac{\sqrt{3}}{2}\right)(x) = 3$$

$$-\sqrt{3}(x) = 3$$

$$x = -\frac{3}{\sqrt{3}}$$

or

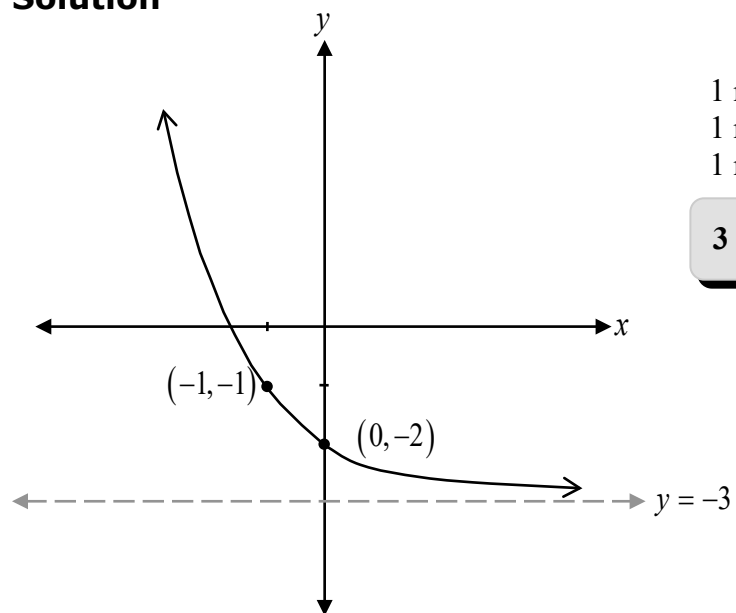
$$x = -\sqrt{3}$$

1 mark for $\sec\left(\frac{2\pi}{3}\right)$ (½ mark for value; ½ mark for quadrant)

1 mark for $\sin\left(-\frac{5\pi}{3}\right)$ (½ mark for value; ½ mark for quadrant)

2 marks

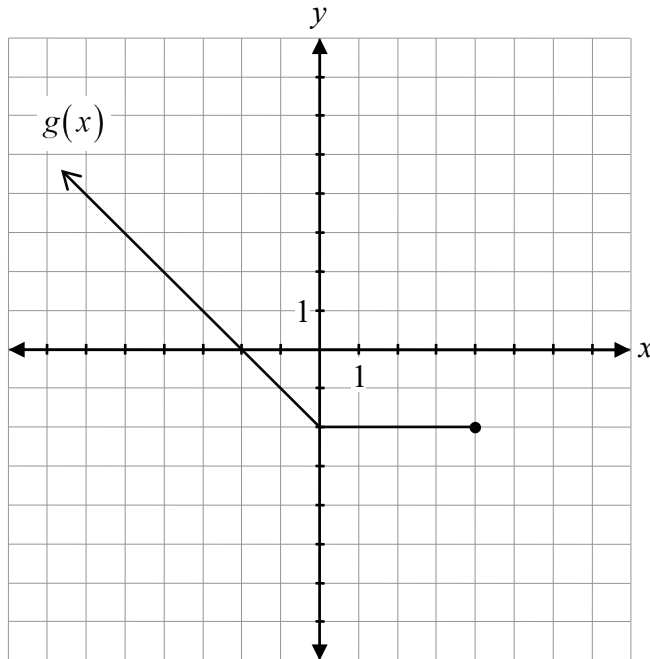
Sketch the graph of $y = 2^{-x} - 3$.

Solution

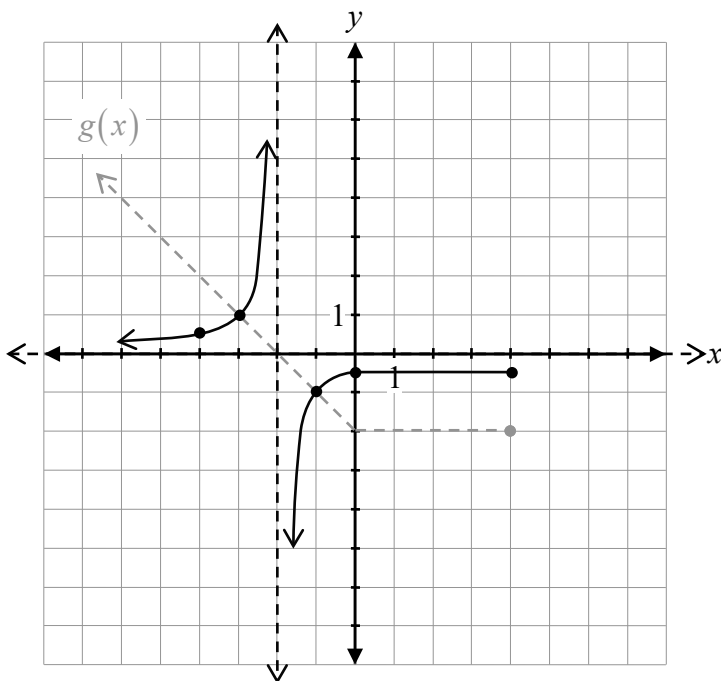
1 mark for shape of an exponential function
1 mark for horizontal reflection
1 mark for vertical translation

3 marks

Given the graph of $y = g(x)$, sketch the graph of $y = \frac{1}{g(x)}$.



Solution



1 mark for asymptotic behaviour approaching $x = -2$ ($\frac{1}{2}$ mark for each side)

$\frac{1}{2}$ mark for asymptotic behaviour approaching $y = 0$

$\frac{1}{2}$ mark for graph left of $x = -2$

$\frac{1}{2}$ mark for graph right of $x = -2$

$\frac{1}{2}$ mark for restricting domain

3 marks

Determine the exact value of $\tan\left(\frac{\pi}{12}\right)$.

Solution

$$\tan\left(\frac{\pi}{12}\right) = \tan\left(\frac{4\pi}{12} - \frac{3\pi}{12}\right)$$

$$\tan\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \frac{\tan\frac{\pi}{3} - \tan\frac{\pi}{4}}{1 + \tan\frac{\pi}{3}\tan\frac{\pi}{4}}$$

1 mark for substitution into correct identity

$$= \frac{\sqrt{3} - 1}{1 + \sqrt{3}}$$

1 mark for exact values (½ mark for each value)

or

$$= 2 - \sqrt{3}$$

or

$$= \frac{3 - \sqrt{3}}{3 + \sqrt{3}}$$

2 marks

Note:

- Other combinations are possible.

Explain why the graph of $g(x) = \frac{3}{x^2 + 4}$ does not have a vertical asymptote.

Solution

The graph of a rational function has a vertical asymptote when the denominator is equal to zero. The denominator of $g(x)$ will never be equal to zero.

1 mark

Solve, algebraically.

$$\log_3 x + \log_3 (x+8) = 2$$

Solution

Method 1

$$\log_3 [x(x+8)] = 2$$

1 mark for product law

$$x^2 + 8x = 3^2$$

1 mark for exponential form

$$x^2 + 8x - 9 = 0$$

$$(x+9)(x-1) = 0$$

$$\cancel{x+9} \quad x = 1$$

½ mark for the permissible value of x

½ mark for showing the rejection of the extraneous root

3 marks

Method 2

$$\log_3 x + \log_3 (x+8) = \log_3 3^2$$

1 mark for logarithmic form

$$\log_3 [x(x+8)] = \log_3 9$$

1 mark for product law

$$x^2 + 8x = 9$$

$$x^2 + 8x - 9 = 0$$

$$(x+9)(x-1) = 0$$

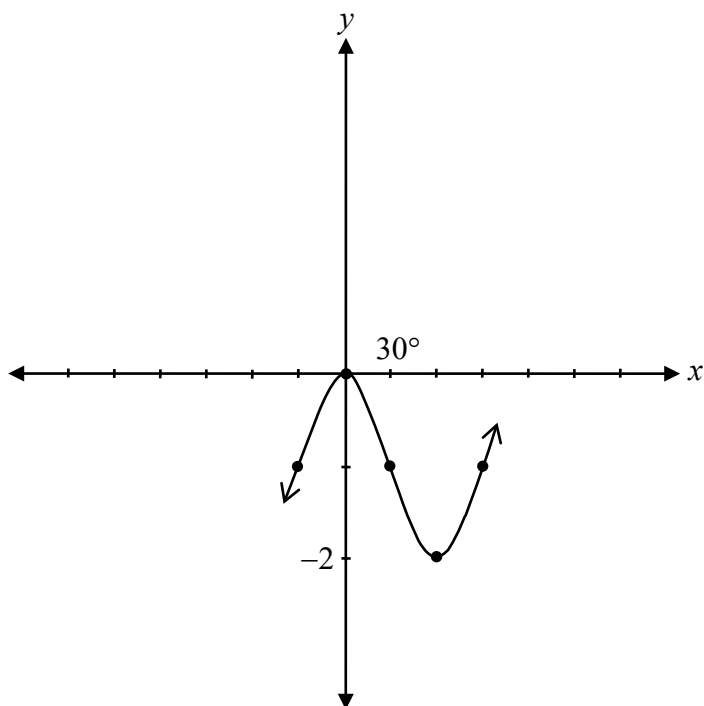
$$\cancel{x+9} \quad x = 1$$

½ mark for the permissible value of x

½ mark for showing the rejection of the extraneous root

3 marks

Sketch at least one period of the graph of the function $y = \sin(3(x + 30^\circ)) - 1$.

Solution

1 mark for shape of a sinusoidal function with correct amplitude

1 mark for period

1 mark for horizontal translation

1 mark for vertical translation

4 marks

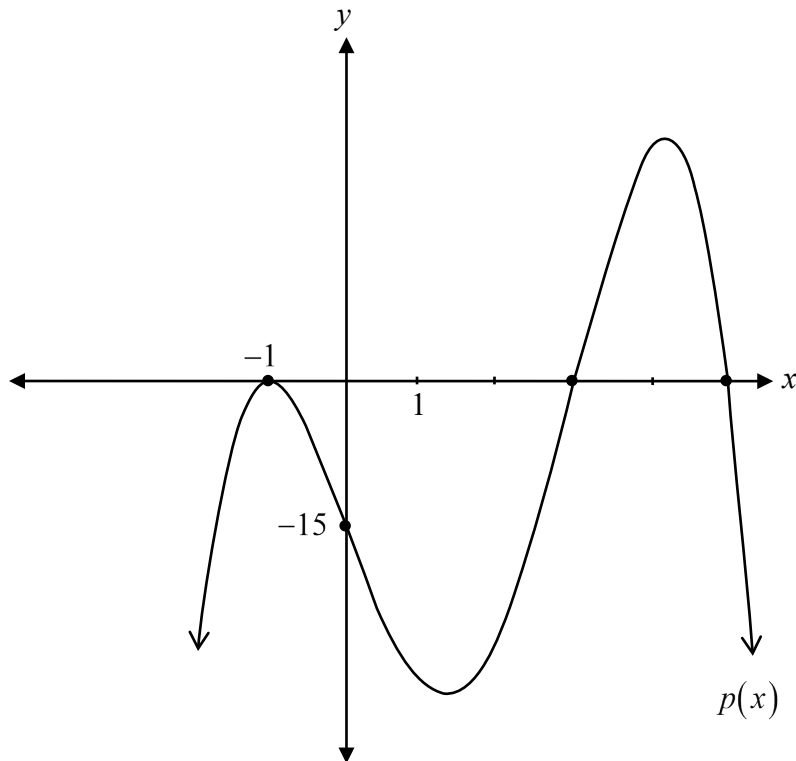
Explain why the domain of the function, $f(x) = \log(x-3)$, is $x > 3$.

Solution

The argument of a logarithm cannot be negative or zero.

1 mark

Sketch the graph of $p(x) = -(x-3)(x+1)^2(x-5)$.

Solution

1 mark for x -intercepts

1 mark for multiplicity of 2 at $x = -1$

$\frac{1}{2}$ mark for end behaviour

$\frac{1}{2}$ mark for y -intercept

3 marks

Given that $\sin \theta = -\frac{2}{3}$ and $\tan \theta > 0$, determine the exact value of $\sin 2\theta$.

Solution

$$x^2 = r^2 - y^2$$

$$x^2 = 9 - 4$$

$$x^2 = 5$$

$$x = \pm\sqrt{5}$$

$\frac{1}{2}$ mark for value of x

$$x = -\sqrt{5}$$

$$\cos \theta = -\frac{\sqrt{5}}{3}$$

$\frac{1}{2}$ mark for value of $\cos \theta$

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$= 2 \left(-\frac{2}{3} \right) \left(-\frac{\sqrt{5}}{3} \right)$$

1 mark for substitution into correct identity

$$= \frac{4\sqrt{5}}{9}$$

2 marks

Note:

- Accept any of the following values for x : $x = \pm\sqrt{5}$, $x = -\sqrt{5}$, or $x = \sqrt{5}$.

Justify whether $\frac{5\pi}{8}$ and $-\frac{11\pi}{4}$ are coterminal angles.

Solution**Method 1**

$$\frac{5\pi}{8} - \left(-\frac{11\pi}{4}\right)$$

$$\frac{5\pi}{8} + \frac{11\pi}{4}$$

$$\frac{5\pi}{8} + \frac{22\pi}{8}$$

$$\frac{27\pi}{8}$$

Since $\frac{27\pi}{8}$ is not a multiple of 2π they are not coterminal angles.

1 mark**Method 2**

Coterminal angles of $\frac{5\pi}{8}$:

$$\frac{5\pi}{8} - \frac{16\pi}{8} = -\frac{11\pi}{8}$$

$$-\frac{11\pi}{8} - \frac{16\pi}{8} = -\frac{27\pi}{8}$$

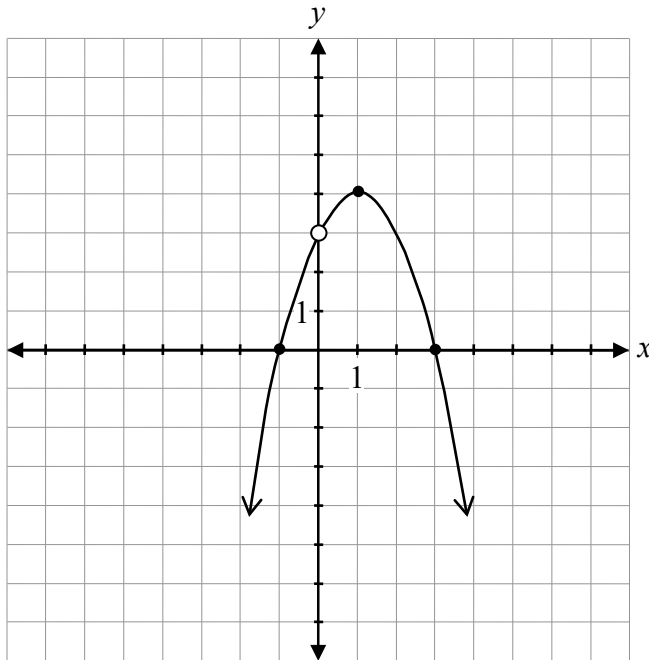
Coterminal angles of $-\frac{11\pi}{4}$:

$$-\frac{11\pi}{4} \left(\frac{2}{2}\right) = -\frac{22\pi}{8}$$

$\therefore$ Angles are not coterminal since $-\frac{27\pi}{8} \neq -\frac{22\pi}{8}$.

1 mark

Sketch the graph of $f(x) = \frac{-2x(x+1)(x-3)}{2x}$.

Solution

1 mark for point of discontinuity (hole) at $x = 0$
½ mark for shape of a parabola with correct
x-intercepts
½ mark for end behaviour

2 marks

Note:

- Deduct ½ mark for procedural error (incorrect y-value for point of discontinuity (hole)).

Given $\frac{\sin \theta + \cos \theta \csc \theta}{\sin \theta}$, determine the non-permissible values of θ , where $\theta \in \mathbb{R}$.

Solution

Method 1

$\sin \theta = 0$ $\frac{1}{2}$ mark for $\sin \theta = 0$
 $\theta = 0, \pi, 2\pi$ $\frac{1}{2}$ mark for any non-permissible value of θ
 $\theta = \pi k, k \in \mathbb{Z}$ 1 mark for all non-permissible values of θ

2 marks

Method 2

$\sin \theta = 0$ $\frac{1}{2}$ mark for $\sin \theta = 0$
 $\theta = 0, \pi, 2\pi$ $\frac{1}{2}$ mark for any non-permissible value of θ
 $\theta = 2\pi k, k \in \mathbb{Z}$ }
 $\theta = \pi + 2\pi k, k \in \mathbb{Z}$ } 1 mark for all non-permissible values of θ

2 marks

Write an equation of a rational function that has a horizontal asymptote at $y = 0$ and a vertical asymptote at $x = 6$.

Solution

$$f(x) = \frac{1}{x-6}$$

1 mark for horizontal asymptote

1 mark for vertical asymptote

2 marks

Note:

- Other answers are possible.

Given the functions $f(x) = \sqrt{x-1}$ and $g(x) = x^2$,

a) state the equation of $g(f(x))$.

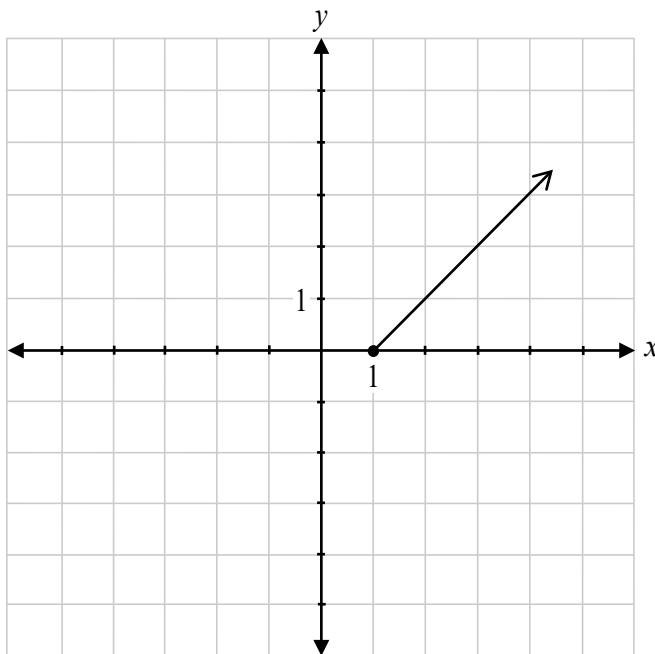
b) sketch the graph of $g(f(x))$.

Solution

a) $g(f(x)) = x - 1, x \geq 1$

1 mark

b)



1 mark for shape of graph consistent with a)
1 mark for restricted domain

2 marks**Notes:**

- Deduct a maximum of 1 mark for the concept error of not restricting the domain.
- Deduct ½ mark for procedural error (not stating domain) if graph shows correct domain.

Suzanne was asked to determine the value of $\tan \theta$, given that $\sec \theta = -\frac{8}{3}$ and θ terminates in quadrant II.

Her solution:

$$\begin{aligned}(-3)^2 + y^2 &= (8)^2 \\ y^2 &= 55 \\ y &= \sqrt{55} \\ \tan \theta &= \frac{\sqrt{55}}{3}\end{aligned}$$

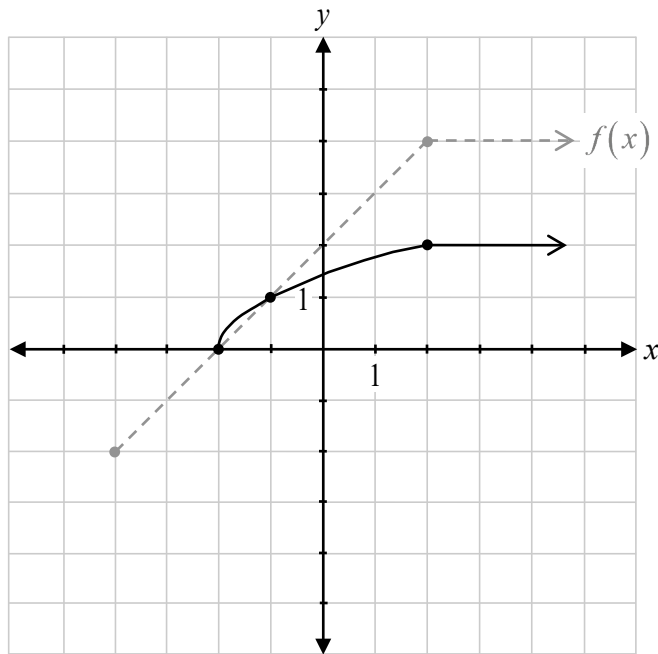
Describe her error.

Solution

Suzanne did not consider that the value of $\tan \theta$ is negative in quadrant II.

1 mark

Given the graph of $y = f(x)$, sketch the graph of $y = \sqrt{f(x)}$.

Solution

1 mark for restricting domain

$\frac{1}{2}$ mark for shape between invariant points

$\frac{1}{2}$ mark for shape to the right of invariant points

2 marks

The point $P(\theta) = (0, -1)$ lies on the unit circle. State the angle θ , over the interval $[2\pi, 4\pi]$.

Solution

$$\theta = \frac{7\pi}{2}$$

1 mark

Describe how the transformations of $f(x)$ on the graphs of $g(x) = f(3x - 6)$ and $h(x) = f(3(x - 6))$ are different.

Solution

The graph of $g(x)$ is a horizontal translation of $f(x)$ two units to the right and the graph of $h(x)$ is a horizontal translation of $f(x)$ six units to the right.

1 mark

a) Solve.

$$\sqrt{2x+5}-3=0$$

b) Describe how the solution in a) relates to the graph of $y = \sqrt{2x+5} - 3$.

Solution

a) $(\sqrt{2x+5})^2 = 3^2$

$$2x+5=9$$

$$2x=4$$

$$x=2$$

1 mark

b) The solution is the x -intercept of the graph.

1 mark

Determine all of the zeros of the function $p(x) = x^3 - 2x^2 - 9x + 18$.

Solution

$$\begin{aligned} p(2) &= 2^3 - 2(2)^2 - 9(2) + 18 \\ &= 0 \end{aligned}$$

1 mark for identifying one possible zero of $p(x)$

$\therefore (x-2)$ is a factor of $p(x)$

$$\begin{array}{r|rrrr} 2 & 1 & -2 & -9 & 18 \\ & & 2 & 0 & -18 \\ \hline & 1 & 0 & -9 & 0 \end{array}$$

1 mark for synthetic division (or equivalent strategy)

$$(x-2)(x^2-9) = 0$$

$\frac{1}{2}$ mark for consistent factors

$$(x-2)(x-3)(x+3) = 0$$

$$x = 2, x = 3, x = -3$$

$\frac{1}{2}$ mark for consistent zeros

3 marks

Given that the point $\left(\frac{\sqrt{23}}{6}, y\right)$ is on the unit circle, determine the exact value(s) of y .

Solution

$$\left(\frac{\sqrt{23}}{6}\right)^2 + y^2 = 1 \quad \frac{1}{2} \text{ mark for substitution}$$

$$\frac{23}{36} + y^2 = 1$$

$$y^2 = \frac{36}{36} - \frac{23}{36}$$

$$y = \pm \frac{\sqrt{13}}{6} \quad \frac{1}{2} \text{ mark for exact values of } y$$

1 mark

State one zero of the function $y = \tan x$.

Solution

Any solution where $x = \pi k$, $k \in \mathbb{Z}$, is acceptable.

1 mark